

Supplementary Material

Supplementary Table 1: Complete highest R^2 expression obtained from the single-objective symbolic regression

R^2 : 0.688275
Complexity: 1995.20
Expression: $((8.29 * (2.71 * \sin(((2.31 * ((\text{Lithology} / 0.09) + (\text{Lithology} / 0.09))) - \sin((2.71 * \exp(((2.71 / (\text{LULC} - \text{Rain})) - (\text{Lithology} - (\text{Lithology} / 0.09)))))) + \exp((\text{Geomorphology} - (\text{Lithology} - \sin((\text{Lineament density} - (\sin((2.71 * \exp((\text{Lineament density} - (\exp(\sin((\text{LULC} - (\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))) - (\text{Lithology} / -9.3)))) + (\text{Lithology} / 0.09)))) - (\text{Lithology} / 0.09)) + \exp((\text{Lineament density} - (\text{Lithology} / 0.09)))))))))) + ((2.71 * ((2.71 * (2.71 / ((\text{LULC} - \text{Rain}) / ((2.31 * (2.71 / \exp((\text{Lineament density} - (\exp(\sin((\text{LULC} - \sin((\text{Geomorphology} * \text{LULC}) - ((2.71 * (2.71 / ((\text{LULC} - \text{Rain}) / ((\text{Lithology} / 0.09) - ((2.71 * (2.71 / ((\text{LULC} - \text{Rain}) / ((2.31 * ((2.31 * ((\text{Lithology} / 0.09) + (\text{Lithology} / 0.09))) + \exp(((2.71 * \sin((\text{Lineament density} - (\text{Lithology} / 0.09)) + \exp((\text{Lineament density} - \exp((\text{Lineament density} - (\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))))))) - (\text{Lithology} / 0.09))) + (\sin((2.71 * \exp((\text{Lineament density} - (2.31 * (((\text{Soil} * \text{Geomorphology}) + (\exp(\sin((\text{LULC} - (\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))) - (\text{Lithology} / -9.3)))) - (\text{Lithology} / -9.3))) * \text{Geomorphology}) + (\text{Lithology} - (\text{Lithology} / 0.09)))))) - (\text{Lithology} / 0.09)) + (\text{Geomorphology} - (\text{Lithology} - \sin((\text{Lithology} / -9.3)))))) - \sin((\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))))) - (\text{Lithology} / -9.3)))) - (\text{Lineament density} - ((2.71 * (\text{Lithology} / -9.3)) + ((2.71 * (2.71 * \sin(((2.31 * (\text{Geomorphology} * \text{LULC})) - \sin((2.71 * \exp(((2.71 / \exp((2.71 / (\text{Lithology} / 0.09)))) - (\text{Lithology} / 0.09)))))) + \exp((\text{Geomorphology} - (\text{Lineament density} - (\sin((2.71 * (\text{Lithology} / -9.3)) + ((2.71 * (2.71 * \sin(((2.31 * ((\text{Lithology} / 0.09) + (\text{Lithology} / 0.09))) - \sin(((2.31 * ((\text{Lithology} / 0.09) + (\text{Lithology} / 0.09))) - \sin((\text{Lineament density} - (\sin((2.71 * \exp((\text{Lineament density} - (\exp((\text{Lineament density} - (\text{Lithology} / 0.09))) + (\text{Lithology} / 0.09)))) + (\text{Lithology} / 0.09)))) - (\text{Lithology} / 0.09)) + \sin(((2.31 * (\text{Geomorphology} * \text{LULC})) - \sin((2.71 * \exp(((2.71 / \exp((2.71 / (\text{Lithology} / 0.09)))) - (\text{Lithology} / 0.09)))))) + \exp((\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))))) + \exp((\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} / 0.09)))))) + \exp((\text{Lineament density} - \exp(\sin((\exp(\sin((\text{LULC} - \sin((\text{Geomorphology} * \text{LULC}) - ((2.71 * (2.71 / ((\text{Lithology} / 0.09) / ((2.31 * ((\text{Soil} * \text{Geomorphology}) + \sin((\text{Rain} / 0.09) - ((2.71 * (2.71 / ((\text{LULC} - \text{Rain}) / ((2.31 * ((\text{Soil} * \text{Geomorphology}) + (\sin((2.71 * \exp((\text{Lineament density} - (2.31 * (((\text{Soil} * \text{Geomorphology}) + (\exp(\exp((\text{Lineament density} - (\text{Soil} + \text{Lithology})))) - (\text{Lithology} / -9.3))) * \text{Geomorphology}) + (\text{Lithology} - (\text{Lithology} / 0.09)))))) - (\text{Lithology} / 0.09)) + (\text{Geomorphology} - (\text{Lithology} - (\text{Lithology} /$

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0.09)))))) - sin((Geomorphology - (Lithology - (Lithology / 0.09))))))
- (Lithology / -9.3)))) - sin((Lithology / -9.3)))) - (Lineament
density - (sin((2.71 * exp((Lineament density - (exp((Lineament density
- (Lithology / 0.09)) + (Lithology / 0.09)))) - (Lithology / 0.09)) +
exp((Lineament density - (Soil * Lithology)))))) - (Lithology /
-9.3)))) - (Lithology / -9.3)))))) + ((2.71 * (Lithology / -9.3)) +
(Geomorphology - (Lithology - (Lithology / 0.09)))))) + exp((Lineament
density - (Lithology / 0.09)))))) + ((2.71 * (Lithology / -9.3)) +
(Geomorphology - (Lithology - (Lithology / 0.09)))))) - (Lithology
/ -9.3)))) + exp((Lineament density - (Soil + Lithology)))))) -
sin((Lithology / -9.3)))) - (Lineament density - (sin((2.71 *
exp((Lineament density - (exp((Lineament density - (Lithology / 0.09))
+ (Lithology / 0.09)))) - (Lithology / 0.09)) + (2.71 * sin(((2.31 *
((Lithology / 0.09) + (Geomorphology * LULC))) - sin((2.71 * exp(((2.71
^ exp((2.71 / (Lithology / 0.09))) - (Lithology / 0.09)))))) +
exp((Geomorphology - (Lithology - (Lithology / 0.09)))))) + ((2.71
* (2.71 * sin(((2.31 * ((Lithology / 0.09) + (Lithology / 0.09)) -
sin((((2.31 * ((Lithology / 0.09) + (Lithology / 0.09)) - sin(((2.31
* ((Lithology / 0.09) + (Lithology / 0.09)) - sin(sin((Geomorphology -
(Lithology - (Lithology / 0.09)))))) + exp((Geomorphology - (Lithology -
(Lithology / 0.09)))))) - sin((2.71 * exp((Lithology / -9.3)))) +
exp((Geomorphology - (Lithology - (Lithology / 0.09)))))) +
exp((Geomorphology - (Lithology - (Lithology / 0.09)))))) + ((2.71 *
(Lithology / -9.3)) + (Geomorphology - (Lithology - (Lithology /
0.09))))))

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